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Compositional Generalization Requires Linear, Orthogonal Representations in Vision Embedding Models

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Abstract

Compositional generalization, the ability to recognize familiar parts in novel contexts, is a defining property of intelligent systems. Although modern models are trained on massive datasets, they still cover only a tiny fraction of the combinatorial space of possible inputs, raising the question of what structure representations must have to support generalization to unseen combinations. We formalize three desiderata for compositional generalization under standard training (divisibility, transferability, stability) and show they impose necessary geometric constraints: representations must decompose linearly into per-concept components, and these components must be orthogonal across concepts. This provides theoretical grounding for the Linear Representation Hypothesis: the linear structure widely observed in neural representations is a necessary consequence of compositional generalization. We further derive dimension bounds linking the number of composable concepts to the embedding geometry. Empirically, we evaluate these predictions across modern vision models (CLIP, SigLIP, DINO) and find that representations exhibit partial linear factorization with low-rank, near-orthogonal per-concept factors, and that the degree of this structure correlates with compositional generalization on unseen combinations. As models continue to scale, these conditions predict the representational geometry they may converge to. Code is available at https://github.com/oshapio/necessary-compositionality.

1. Introduction

Modern vision systems are trained on a tiny, biased subset of a combinatorial space of visual concepts, like objects, attributes, relations in different contexts. Despite this, we expect them to perform well in the wild on novel recombinations of familiar concepts, an expectation tied to the view that systematic generalization, the ability to recombine learned constituents, is a hallmark of intelligence (Fodor & Pylyshyn, 1988). Yet a large body of empirical work shows that even high-performing neural models often struggle with systematicity when train/test combinations mismatch (Lake & Baroni, 2018; Keysers et al., 2020; Hupkes et al., 2022; Uselis et al., 2025). At the same time, large vision-language models such as CLIP (Radford et al., 2021) and its variants are trained on web-scale datasets (e.g., LAION-400M (Schuhmann et al., 2021a)) and achieve impressive zero-shot transfer on many tasks (Radford et al., 2021; Zhai et al., 2022). However, these systems often fail when test images contain unusual combinations of familiar concepts (Xu et al., 2022; Bao et al., 2024; Thrush et al., 2022; Abbasi et al., 2024; Yuksekgonul et al., 2023; Ma et al., 2023). Fig. 1 illustrates this tension for CLIP-like architectures: an image encoder f produces embeddings on which linear classifiers (e.g. from the text encoder) classify concepts, but the training data $\mathcal{X}_{\text{train}}$ cover only common compositions (such as a cat on a person) from the full data space $\mathcal{X}$, while models must answer queries like "Is there a person present?" correctly even on rare compositions (such as a cartoon of a person on a cat) from $\mathcal{X} \setminus \mathcal{X}_{\text{train}}$. Given how rarely, if at all, such compositions appear in training, we ask: *assuming that compositional generalization succeeds, what properties must the representations have to accommodate it?*

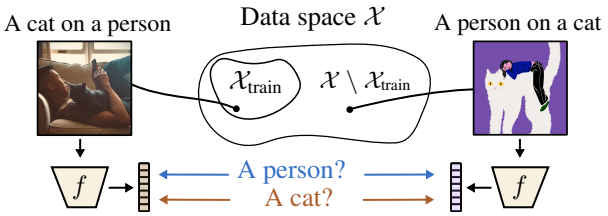

Figure 1. **What enables compositional generalization in vision-language embedding models?** Training data contain common configurations (left: a cat on a person) but lack rare ones (right: a person on a cat). Yet the same text-based queries, e.g. "A photo of a person", must work on both, even when the latter was never seen during training. We investigate what properties encoder f must satisfy for such transfer to succeed.

We argue for *non-negotiable, model-agnostic* properties that

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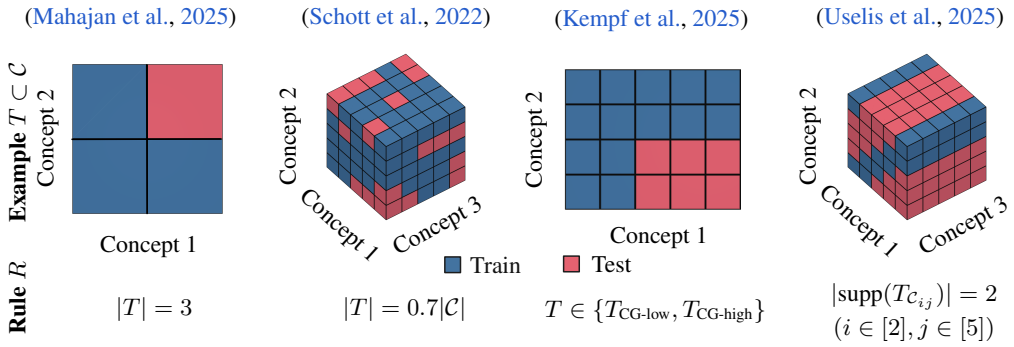

Figure 2. **Interpreting previous works' sampling designs T and validity rules R.** Each panel shows a concept space as a grid (2D or 3D), with blue cells denoting training combinations and red cells denoting held-out test combinations.

any neural-network-based system claiming compositional generalization must satisfy. We state three desiderata: *divisibility*, *transferability*, and *stability*. These desiderata formalize that (i) all parts of an input should be accessible to a simple readout; (ii) readouts trained on a tiny but diverse subset should transfer to unseen combinations; and (iii) training on any valid subset should yield robust generalization. Our scope is the common setting where predictions are linear in the embedding f: CLIP-style zero-shot classifiers, linear probing, and cases where a fixed non-linear head is folded into the encoder.

Our key finding is that these desiderata *necessitate* a specific geometry: *linear factorization* with *near-orthogonal* concept directions. This establishes what any model *must* achieve to compositionally generalize under standard training, providing a concrete target for future design. Moreover, it offers theoretical grounding for the *Linear Representation Hypothesis* (Mikolov et al., 2013; Elhage et al., 2022) – the linear structure widely observed in neural representations is a *necessary consequence* of compositional generalization.

Our contributions are: (1) **Defining desiderata.** We define three desiderata: *divisibility*, *transferability*, *stability*, and formalize compositional generalization in their terms. (2) **Structural necessity.** Under gradient descent with the cross-entropy loss, these desiderata imply *linear factorization*: embeddings decompose into per-concept sums with orthogonal difference directions. (3) **Empirical grounding.** Across CLIP and SigLIP families, we find strong evidence of factorization, near-orthogonality, low-rank per-concept geometry, and correlation with compositional generalization accuracy, suggesting that the current state-of-the-art vision models are converging to the necessary geometric properties.

2. Related work

Compositional generalization. Prior work establishes *sufficient* conditions for compositional generalization under specific assumptions on data-generating processes or representations (Wiedemer et al., 2023; Mahajan et al., 2025; Uselis et al., 2025). In contrast, we study *necessary* conditions: we ask what must be true of a model's embeddings *if* it transfers from a restricted subset to the full space under our desiderata.

Geometry of learned representations. Empirical studies document linear subspaces in VLMs (Trager et al., 2023), the Linear Representation Hypothesis in LLMs (Park et al., 2023), and near-orthogonal feature encodings (Elhage et al., 2022). In parallel to our work, (Fel et al., 2025) map task-relevant concept families in DINOv2 and propose a Minkowski-sum hypothesis for token space, highlighting that strong geometric structure may arise from architecture as well as from downstream demands. In contrast, we show that under practice-driven desiderata and standard training, linearity and orthogonality are *necessary*, not merely observed.

Disentangled and object-centric representations. This line of work proposes desiderata and training schemes, with mixed evidence linking these to compositional generalization (Eastwood & Williams, 2018; Montero et al., 2021; Dittadi et al., 2022). We ask the complementary question: if a model *does* generalize compositionally, what must its embeddings satisfy? See Section B for a detailed discussion.

3. Setup: A Framework for compositionality

We begin by detailing key desiderata for embedding models that contend to be compositionally generalizing. We motivate them from a practical perspective: (1) models need to support distinguishing between any combination of concepts, (2) practical data collection is limited to a subset of the concept space, so a model needs to be able to transfer from a subset of the concept space to the full concept space, and (3) in practice apriori it is not known which subset needs to be chosen, so a model should be able to transfer robustly from any subset, matching in probability distribution to retraining over any other dataset.

3.1. Setup: concept spaces and data collection process

We interpret the world as a product of concepts: any input $\boldsymbol{x_c} \in \mathcal{X}$ (e.g., an image) has an associated tuple of concepts $\boldsymbol{c} \in \mathcal{C}$, describing its constituent parts and properties. This is a reasonable way to describe a large portion of the world. For example, current large-scale datasets (e.g., image-caption pairs) provide noisy natural-language descriptions that can be decomposed into *discrete* concept values. Clearly, a single concept tuple cannot capture all aspects of the world, e.g. how attributes bind to objects or how different objects relate spatially. Still, an intelligent system should at least be able to tell apart basic concepts (such as objects and their attributes), even without modeling their relations. In other words, concept spaces may not capture the full compositional structure of the world, but any model of the world must involve them in some form. Importantly, we do not assume *how* the concept values are distributed (e.g. being independent), only *what* they represent.

Definition 1 (Concept space)**.** Suppose we have k concepts, where concept i takes n_i possible values. Let $\mathcal{C}_i = [n_i]$ for $i \in [k]$. The *concept space* is the Cartesian product

$$\mathcal{C} = \mathcal{C}_1 \times \mathcal{C}_2 \times \cdots \times \mathcal{C}_k. \tag{1}$$

In most of the paper we use $n_i = n$ for all i, so $\mathcal{C} = [n]^k$ and $|\mathcal{C}| = n^k$. We index inputs by concept tuples: for each $\boldsymbol{c} \in \mathcal{C}$ we assume an associated $\boldsymbol{x_c} \in \mathcal{X}$ (e.g., a natural image) realizing $\boldsymbol{c}$.

Data-related components for compositional generalization involve three notions: (1) the total variation of the data, (2) the concepts we aim to learn and expect the model to capture, and (3) the data that is actually collectible. We capture (1) by the concept space $\mathcal{C}$ (Definition 1). For (2), the targets that we aim to capture can be described by a label function $l : \mathcal{C} \to \mathcal{V} \subseteq \mathcal{C}$ that captures which concepts and their values we want to learn.[1] In this work we take the full target $\mathcal{V} = \mathcal{C}$, reflecting that foundation models attempt to align with all present concepts. For (3), we formalize collectability constraints through a validity class that specifies which training supports are valid, indicating which concept combinations may appear in training–taking into account cases that many combinations are not collectible (e.g. "a pink cat on the moon" isn't common). We formalize this below.

Considering data collection. We are interested in models that support efficient compositional generalization from a subset of the concept space. To formalize this notion, we specify a validity class $\mathcal{T} \subseteq 2^{\mathcal{C}}$ of valid training sets, where $2^{\mathcal{C}}$ denotes the power set of $\mathcal{C}$, and a validity rule $R : 2^{\mathcal{C}} \to \{0, 1\}$ that specifies whether a given training set is valid. This setup captures the natural question of which training sets we use and for which we expect generalization.

[1] More generally, the world may involve additional factors beyond the concepts represented in $\mathcal{C}$. Our framework does not require $\mathcal{C}$ to be exhaustive: $\mathcal{C}$ can be viewed as the subset of concepts we model explicitly, with any remaining variation treated as nuisance.

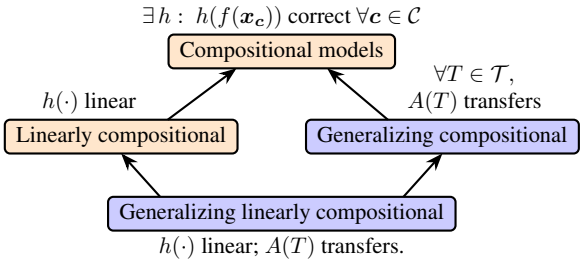

Figure 3. **Relationship between (generalizing) compositional models.** Divisibility (orange) and transferability (blue) requirements.

Definition 2 (Training support, validity class, and training dataset)**.** Let $\mathcal{C}$ be the concept space. A *training support* is any subset $T \subseteq \mathcal{C}$. *Validity class* is a collection $\mathcal{T} \subseteq 2^{\mathcal{C}}$ whose members are called *valid training sets*. The class $\mathcal{T}$ specifies which training sets are observable. Validity class $\mathcal{T}$ is specified by a *validity rule* $R : 2^{\mathcal{C}} \to \{0, 1\}$ through $\mathcal{T} = \{T \subseteq \mathcal{C} : R(T) = 1\}$. A *training dataset* for a training set T is $D_T = \{(\boldsymbol{x_c}, \boldsymbol{c}) : \boldsymbol{c} \in T\}$.

We note that there are many validity rules used in practice. For example, if we can collect any subset of size $N < |\mathcal{C}|$, then $R(T) = 1$ whenever $|T| = N$. Fig. 2 illustrates common choices: Mahajan et al. (2025) use training supports that cover every concept value; Schott et al. (2022) use random samples covering 70% of all combinations; Kempf et al. (2025) specify a small set of allowed supports; and Uselis et al. (2025) use supports whose joint marginals cover at least two values per concept. Note that these validity rules apply to concept supports rather than individual datapoints.

3.2. Compositional representations and models

Given the concept space and the training supports, we now make precise how we expect models to learn. We work with encoders f that map an input to a vector representation (embedding).

Scope of models. We study embedding models: these cover modern foundation models like CLIP and SigLIP (Tschannen et al., 2025; Zhai et al., 2023), supervised-learning models, self-supervised models like DINO (Caron et al., 2021; Siméoni et al., 2025). At inference the models we study are *non-contextual*: the representation of an input depends only on that input (no dependence on other test examples, prompts, or the batch). Formally, the encoder is a map $f : \mathcal{X} \to \mathcal{Z}$, with $\boldsymbol{z} = f(\boldsymbol{x})$ (optionally ℓ_2-normalized).

Readout class (linear vs. non-linear). Usually, encoders f are associated with either a downstream or readout model h that takes $\boldsymbol{z} = f(\boldsymbol{x})$ and outputs per-concept logits $h(\boldsymbol{z}) \in \mathbb{R}^{k \times n}$ using argmax classification rule (see Definition 3).